STAFF REVIEW · H446 137 source models

Separate from the eight-page pupil workbook. This copy is in a public repository and is NOT access controlled. Check technical correctness before use.

Printed question mapping: Booklet Q1 = original online Q1; Booklet Q2 = original online Q2; Booklet Q3 = original online Q3

Do Now — source answers

1. A non-key attribute depends on only part of a composite candidate key.

2. Read-only memory.

3. Sum 0, carry 1.

4. A key made from two or more attributes together.

I Do — source worked steps

1. EmpID determines DeptID — Find the given data and name the required result.

2. DeptID determines name — Explain why this intermediate stage is necessary.

3. Split Department — EmpID identifies employee. DeptName depends on DeptID, not directly on EmpID, producing a transitive dependency. Split Employee(EmpID PK,EmpName,DeptID FK) and Department(DeptID PK,DeptName).

We Do — guided CHECK

1. Order identifies customer
Find the given data and name the required result.
Customer identifies postcode
Explain why this intermediate stage is necessary.
Split or preserve snapshot
Under the assumption CustomerID determines the customer's current postcode, there is an indirect dependency OrderID→CustomerID→CustomerPostcode. Store Customer(CustomerID PK,Postcode) and Order(OrderID PK,CustomerID FK). Historical shipping addresses may need separate, deliberate snapshots.

You Do — source mark schemes

1. Indirect key→non-key→other non-key chain [3], suitable accurate example [3].

2. Device(DeviceID PK,RoomID FK) [3], Room(RoomID PK,RoomName) [3].

3. Avoid duplicate facts [2], update/insert/delete benefits [2], joins across tables [2], qualified performance and context [2].

Exit Ticket — source answer

1. False. Repeated dependent data can create anomalies.